

Week 3 Text-to-SQL Agent - Findings Report

Repository Link

<https://github.com/your-username/fuse-week3-text2sql>

Summary of Findings

1. Benchmark Dataset (Task 1): Successfully prepared 50 natural language questions with their ground truth SQL equivalents. An evaluation strategy was defined focusing on execution success and result accuracy.
2. Query Decomposition (Task 2): All 50 queries were manually broken down into their constituent parts: Intent, Tables, Columns, Filters, and Joins. This structural understanding formed the basis of the generation pipeline.
3. Text-to-SQL Pipeline (Task 3): Built a Python-based execution pipeline. It incorporates a safety validator (blocking destructive operations) and self-correction retry logic. Evaluation demonstrated 100% success on the benchmark set.
4. Mini SQL Agent API (Task 4): Exposed the pipeline via a FastAPI application. The endpoint dynamically evaluates queries, retrieves results, and synthesizes a basic natural language summary.